Ligjerata 5c

Lëvizja e qendrës së masës së sistemit të izoluar të trupit

$$x_c = \frac{\sum_{i=1}^{n} m_i x_i}{\sum_i m_i}$$

$$\frac{dx_c}{dt} = \frac{\sum_{i=1}^{n} m_i \frac{dx_i}{dt}}{\sum_i m_i}$$

$$v_{xc} = \frac{\sum_{i=1}^{n} m_1 v_{xi}}{\sum_i m_i}, \quad v_{yc} = \frac{\sum_{i=1}^{n} m_i v_{yi}}{\sum_i m_i}, \quad v_{zc} = \frac{\sum_{i=1}^{n} m_i v_{zi}}{\sum_i m_i}$$

$$\vec{v}_c = \frac{\sum_{i=1}^{n} m_i \vec{v}_i}{\sum_i m_i}$$

$$\sum_{i=1}^{n} m_i \vec{v}_i = \text{konst}, \quad \sum_i m_i = \text{konst.}$$

$$\vec{v}_c = \text{konst.} \qquad mv = p$$

$$a_c = \frac{dv_c}{dt} = \frac{1}{m} \frac{dp}{dt} \rightarrow \text{nxitimi i qendrës së masës.}$$

$$F_c = \frac{dp}{dt}$$

$$a_c = \frac{F_c}{m} \rightarrow \boxed{\vec{F}_c = m\vec{a}_c}$$